

Mitchell Oriahi

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EDUCATION

Texas Tech University, Lubbock, TX

Expected May 2027

B.S. Computer Engineering | GPA: 3.46/4.00 | Accelerated M.S. EE track | President's and Dean's Honor Lists

Relevant Coursework: Embedded Systems, Data Structures and Algorithms, Microcontrollers, FPGA Project Lab, Control Systems

PROJECTS

APRS GPS Tracking System

Team Lead, TTU ECE 3334 Project Lab

January 2026 – May 2026

- Led the design of a GPS tracker on two STM32 NUCLEO-G431RB boards in bare-metal C that parses GPS sentences over UART and packs position data into APRS frames with HDLC framing, bit stuffing, NRZI, and CRC-16
- Implemented a Bell 202 AFSK modem in firmware: a timer ISR steps a 256-entry sine table into the DAC for 1200 baud transmit; the receiver samples the ADC and recovers bits with a Goertzel filter and a 32-phase timing bank
- Diagnosed inconsistent frame decoding with oscilloscope captures and bit-level UART logs, isolated receive-path noise and a failed transceiver, and identified input filtering and staged loopback tests as next-revision fixes

STM32 WAV Audio Player

Final Project, TTU ECE 4380 Embedded Systems

January 2026 – May 2026

- Built a FreeRTOS music player on an STM32F429I Discovery board with a TouchGFX touchscreen UI that browses WAV files on an SD card over SDIO with FatFS and supports play, pause, skip, seek, shuffle, and elapsed time
- Kept playback gap-free with a double-buffered DMA pipeline: a hardware timer paces the DAC at the file sample rate while DMA half and full transfer interrupts signal the audio task to refill the idle half
- Split the firmware into four FreeRTOS tasks with a mutex guarding FatFS, keeping the touchscreen responsive during playback and recovering from a card pulled mid-song by force-resetting the SDIO peripheral and remounting

EXPERIENCE

Texas Tech Nano Tech Center, Lubbock, TX

Undergraduate Research Assistant

January 2026 – May 2026, August 2026 – Present

- Designed photolithography masks in AutoCAD and fabricated gold test structures through resist patterning, etching, and deposition to measure resistance versus voltage and temperature
- Characterized aluminum nitride films grown on silicon with X-ray diffraction alongside a graduate researcher, reading diffraction peaks to confirm film presence and orientation before deciding whether to repeat a growth
- Brought up an SP-150 potentiostat for impedance spectroscopy on known test circuits, writing a Python script that overlays measured impedance on the circuit's theoretical response

Christian Brothers Automotive Corporate, Katy, TX

Software Engineer Intern, SDET

May 2026 – August 2026

- Built a canary release system in Python and Azure CLI that deploys to an App Service deployment slot, ramps live traffic from 0 to 25 percent with slot traffic routing, and alerts the deployer; now standard across 7 services
- Automated regression testing in Azure DevOps: publishing a shared library wheel to Azure Artifacts triggers pipelines that deploy 5 consuming services with it pinned, wait for healthy startup, and test all 48 endpoints
- Migrated 4 production services and APIs from Python 3.10 to 3.13 ahead of an Azure runtime deprecation; also built a daily self-updating documentation site covering 46 Azure DevOps projects

Christian Brothers Automotive Corporate, Katy, TX

Data Engineering Intern

May 2025 – August 2025

- Replaced a manual ingestion process with a Python and Azure Functions pipeline, scheduled weekly by a Logic App, that pulls new vehicle records from REST APIs into MySQL and Cosmos DB for the scheduler's vehicle dropdowns
- Cut redundant database writes by over 90 percent with change-detection logic that inserts only new or modified records, and validated API responses before insert so incomplete data never reached the scheduler
- Decoupled fetching from writing with Azure Storage Queues, one message per record, with failed writes routed to a poison queue for retry; deployed through dev, staging, and production and monitored in Application Insights

TECHNICAL SKILLS

Languages: C, Python, Java, Verilog, Assembly (AVR), JavaScript, HTML, CSS

Embedded: FreeRTOS, bare-metal register programming, DMA, interrupts, device drivers, STM32 HAL, TouchGFX, FatFS

Hardware and Protocols: STM32, MSP430, UART, SPI, I2C, SDIO, AFSK, PWM, PCB design, oscilloscope, LTspice, Vivado

Cloud and Tools: Azure (Functions, CLI, Logic Apps, DevOps, Artifacts, App Insights), Git, pytest, MySQL, Cosmos DB, Linux

Lab: AutoCAD mask design, photolithography, X-ray diffraction, Profilm3D profilometry, SP-150 potentiostat (EIS)